

Mathematical and Computational Methods

Exercise Sheet 7: Differentiation I: limits and the rules

Tutorial route (►, in this order, about 80 minutes): 1.2(b,c), 2.6, 2.7, 2.8, 2.9, 2.4(b), 2.1(a). Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Limits, continuity

Exercise 1.1 (★ drill limits at a point). Evaluate, cancelling a common factor or rationalising first where needed:

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2};$

(b) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 - x - 6};$

(c) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x};$

(d) $\lim_{x \rightarrow 0} \frac{\sin 5x}{x}.$

Exercise 1.2 (► ★ drill one-sided limits and asymptotes). (a) Find $\lim_{x \rightarrow 0^+} \frac{1}{x}$ and $\lim_{x \rightarrow 0^-} \frac{1}{x}$, and state the vertical asymptote.

(b) For $f(x) = \begin{cases} x^2, & x < 1 \\ 2x, & x \geq 1 \end{cases}$ find the one-sided limits at $x = 1$ and decide whether $\lim_{x \rightarrow 1} f(x)$ exists.

(c) Divide by the highest power of x in the denominator to find $\lim_{x \rightarrow \infty} \frac{2x+1}{x^2+3}$ and $\lim_{x \rightarrow \infty} \frac{4x^3-x}{2x^3+7}$, and state each horizontal asymptote $y = L$.

Exercise 1.3 (★ drill continuity). Find the constant that makes each function continuous everywhere.

(a) $f(x) = \frac{x^2 - 1}{x - 1}$ for $x \neq 1$, and $f(1) = k$.

(b) $g(x) = \begin{cases} cx + 1, & x \leq 2 \\ x^2, & x > 2. \end{cases}$

The derivative; standard derivatives and rules

Exercise 1.4 (*★ drill first principles*). Using $f'(x) = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$, differentiate from first principles: (a) $f(x) = x^2 - 3x$; (b) $f(x) = \frac{1}{x}$.

Exercise 1.5 (*★ drill standard derivatives, successive derivatives*). Write down the derivative of each (use a power where helpful):

$$(a) x^5, \quad (b) \sqrt{x}, \quad (c) \frac{1}{x^3}, \quad (d) e^{2x}, \quad (e) \ln x, \\ (f) \sin 3x, \quad (g) \cos \frac{x}{2}, \quad (h) \tan x, \quad (i) \cosh x.$$

(j) For $y = x^4$, list y' , y'' , y''' , $y^{(4)}$, $y^{(5)}$.

Exercise 1.6 (*★ drill product and quotient rules*). Name u and v each time. Product rule $(uv)' = u'v + uv'$: (a) $x^2 e^x$; (b) $x \sin x$. Quotient rule $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$: (c) $\frac{x}{x+1}$; (d) $\frac{\sin x}{x}$.

Exercise 1.7 (*★ drill chain rule*). Differentiate: (a) $(3x+1)^5$; (b) e^{x^2} ; (c) $\sin(x^2)$; (d) $\ln(1+x^2)$.

Further techniques

Exercise 1.8 (*★ drill parametric, implicit, logarithmic drill*). Find $\frac{dy}{dx}$: (a) parametrically, for $x = \cos t$, $y = \sin t$; (b) implicitly, for $xy = 1$; (c) logarithmically, for $y = x^x$, $x > 0$.

2 Tutorial problems

Exercise 2.1 (*► ★★ core continuity, differentiability and a corner*). Both parts turn on the one-sided limits of the difference quotient at the join.

- (a) Let $f(x) = \begin{cases} x^2, & x \leq 1 \\ ax + b, & x > 1. \end{cases}$ Find a, b so that f is differentiable at $x = 1$. Why is your f automatically continuous there?
- (b) Using the one-sided limits of the difference quotient at 0, show that $\text{ReLU}(x) = \max(0, x)$ is continuous at 0 but not differentiable there.

Exercise 2.2 (*★★ core first principles and the tangent (geometric reading)*). Let $f(x) = \sqrt{x}$.

- (a) Show from first principles that $f'(x) = \frac{1}{2\sqrt{x}}$.
- (b) Hence find the equation of the tangent line to $y = \sqrt{x}$ at $x = 4$.

Exercise 2.3 (*★★ core velocity and acceleration — physics*). A particle moves along a line with displacement $s(t) = t^3 - 6t^2 + 9t$ metres, $t \geq 0$ seconds.

- (a) Find the velocity $v(t) = \dot{s}$ and acceleration $a(t) = \ddot{s}$.
- (b) When is the particle momentarily at rest?
- (c) In which direction is it moving at $t = 2$?
- (d) Give the acceleration at each rest instant and interpret its sign.

Exercise 2.4 (► ★★ *core sensitivity, the sigmoid and a learning curve — data science*). (a) The logistic function is $\sigma(x) = \frac{1}{1 + e^{-x}}$. Show, using the chain rule, that $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

(b) A model's error is $L(w) = (w - 3)^2 + 1$. Compute $\frac{dL}{dw}$ and evaluate it at $w = 5$. Reading the derivative as a *sensitivity*, should a *small* gradient-descent step increase or decrease w ? Write out the gradient-descent update $w \leftarrow w - \eta \frac{dL}{dw}$ (learning rate $\eta > 0$) for this L . (Unit 8 develops the method, including how large η may be.)

(c) During training a loss decays as $\ell(t) = e^{-t/5}$, where t is the epoch number. Find the rate of change $\ell'(t)$, evaluate it at $t = 0$ and $t = 5$, and say what the two values tell you about training.

Exercise 2.5 (★★ *core spot the error — conceptual*). Each line contains one mistake. Identify it and give the correct derivative.

(a) “ $\frac{d}{dx}(x^2 + 1)^3 = 3(x^2 + 1)^2$.”

(b) “ $\frac{d}{dx}(x^2 e^x) = 2x e^x$.”

(c) “ $\frac{d}{dx}(\sin x \cos x) = \cos x (-\sin x)$.”

Exercise 2.6 (► ★★ *core quotient-then-chain (worked-example anchor)*). Differentiate $y = \frac{(2x - 1)^4}{x^2 + 1}$, simplifying as far as possible.

Exercise 2.7 (► ★★ *core implicit differentiation (worked-example anchor)*). The folium of Descartes is $x^3 + y^3 = 6xy$.

(a) Find $\frac{dy}{dx}$ by implicit differentiation.

(b) Hence find the slope of the curve at the point $(3, 3)$.

Exercise 2.8 (► ★★ *core parametric tangents*). A curve is given by $x = t^2$, $y = t^3 - 3t$.

(a) Find $\frac{dy}{dx}$ in terms of t .

(b) Find the point(s) with a horizontal tangent and the point(s) with a vertical tangent.

Exercise 2.9 (► ★★ *core logarithmic differentiation — a multi-factor expression*). Use logarithmic differentiation to differentiate $y = (x^2 + 1)^3 (2x - 1)^4 e^{-5x}$.

3 Further practice (home)

Exercise 3.1 (★★ *core classify the breaks*). For each function, find every point where the graph breaks (the function is undefined there, or is defined but not continuous) and classify the break as *removable* (the two one-sided limits exist and agree, but the function is undefined there or takes a different value), a *jump* (both one-sided limits exist but differ), or *infinite* (a one-sided limit is $\pm\infty$, so the graph has a vertical asymptote).

$$(a) f(x) = \frac{x - 2}{x^2 - 4}, \quad (b) g(x) = \frac{1}{x - 1}, \quad (c) h(x) = \begin{cases} x, & x < 0 \\ x + 2, & x \geq 0. \end{cases}$$

Exercise 3.2 (★★ *core true or false — conceptual*). State whether each is true or false, with a one-line reason.

(i) If f is continuous at a then f is differentiable at a .

- (ii) $(uv)' = u'v'$.
- (iii) For $\lim_{x \rightarrow a} f(x)$ to exist, $f(a)$ must be defined.
- (iv) If $\lim_{x \rightarrow \infty} f(x) = L$ then $y = L$ is a horizontal asymptote.
- (v) $\frac{d}{dx}x^n = nx^{n-1}$ holds only for integer n .
- (vi) $f(x) = x^{1/3}$ is differentiable at $x = 0$.

Exercise 3.3 (★★core the differentiation bank). Differentiate (state the rule used):

(a) $e^{x^2} \sin x$, (b) $x^2 \ln x$, (c) $\frac{x}{\sqrt{x^2 + 1}}$, (d) $e^{-x} \cos 2x$, (e) $\frac{\ln x}{x}$, (f) $\sqrt{1 + \sqrt{x}}$ ($x > 0$).

Exercise 3.4 (★★core parametric bank, including a second derivative). (a) For $x = 2t$, $y = t^2 - 1$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(b) For the circle $x = r \cos t$, $y = r \sin t$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Exercise 3.5 (★★core implicit bank, including a second derivative). (a) Find $\frac{dy}{dx}$ for $x^2 - xy + y^2 = 7$.

(b) Find $\frac{dy}{dx}$ for $\sin(xy) = x$.

(c) For the hyperbola $x^2 - y^2 = 16$, find $\frac{dy}{dx}$ and then $\frac{d^2y}{dx^2}$, expressing the second derivative purely in terms of y .

Exercise 3.6 (★★core logarithmic differentiation of a quotient). Use logarithmic differentiation to differentiate $y = \frac{\sqrt{x^2 + 1} e^x}{(x + 3)^4}$.

Exercise 3.7 (★★★challenge harder limits). Evaluate, quoting $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$ where needed:

(a) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$, (b) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$, (c) $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$.

Exercise 3.8 (★★★challenge the cycloid). A cycloid is traced by $x = a(t - \sin t)$, $y = a(1 - \cos t)$ ($a > 0$, $t \geq 0$). Show that $\frac{dy}{dx} = \cot \frac{t}{2}$, and state the values of t at which the tangent is horizontal.

Exercise 3.9 (★★★challenge a tower of powers). Differentiate $y = x^{x^x}$ for $x > 0$. Hint: take logs twice, or differentiate x^x first.